

RAMAIAH INSTITUTE OF TECHNOLOGY
DEPARTMENT OF CIVIL ENGINEERING

Course Code: ESC 231

Time allotted: 1Hr

CIE : I

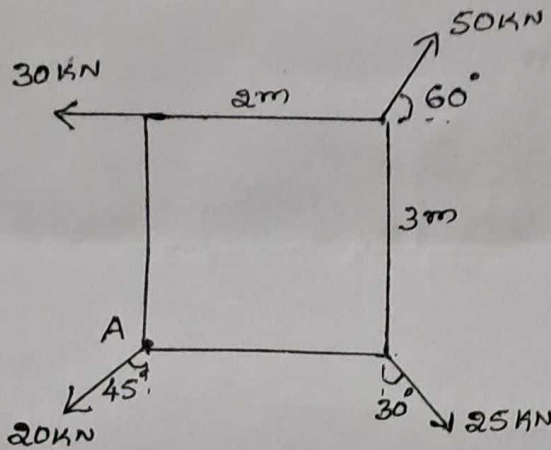
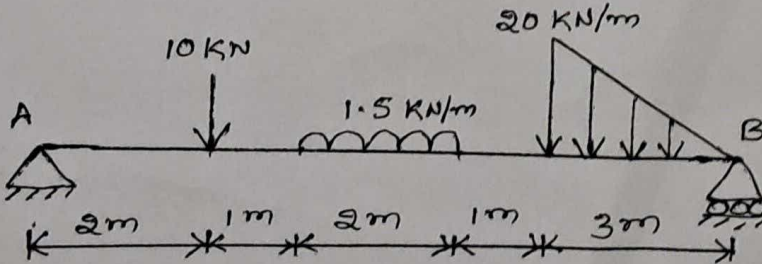
Course: Introduction to Civil
Engineering
Max Marks - 30

Credits: 3:0:0

Semester: 2nd Sem

Term : March – June 2024

Note : Answer any two full questions.

Sl No		Questions	Marks	COs	Blooms
1	a	Explain the role of foundations in building structural system.	5	1	L2
	b	Explain briefly following fields of civil engineering. i) Structural Engineering ii) Geotechnical Engineering.	10	1	L2
2	a	State and prove Varignon's theorem.	5	3	L2
	b	Find the magnitude, direction and point of application of resultant from point 'A' shown in figure 2b. 	10	3	L3
		Figure 2b			
3	a	State the conditions of static equilibrium for coplanar concurrent and non-concurrent force system.	5	3	L2
	b	Find the support reaction for the beam shown in figure 3b. 	10	3	L3
		Figure 3b			